

Caleb A. James

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EDUCATION

George Mason University

Intended B.S., Information Technology, Cybersecurity GPA: 3.65 (Weighted), Honors College

June 2025 – Present

Virginia Polytechnic Institute and State University

Intended B.S., Computer Science, GPA: 3.55 (Weighted), Honors College

August 2024 – June 2025

WORK EXPERIENCE

Undergraduate Researcher

Virginia Tech – IDEEAS Labs

November 2024 – May 2025

- Co-Developed an AI-powered web database using Python, BeautifulSoup, and Hugging Face to collect, format, and organize AI policy data from 400+ higher education institutions, culminating in an AI-powered national policy database.
- Helped maintain and structure a large collaborative policy database while extensively evaluating a wide range of AI models, providers, and companies, identifying differences in model capabilities, extraction, and analysis.
- Contributed to research resulting in a 150+ page academic paper currently undergoing publication, synthesizing findings from the project and documenting the methods used to collect, process, and analyze AI policies across U.S. higher education.

Web Developer

George Mason – The Forge Magazine

September 2026 – Present

- Developed multiple websites for student publication with PHP, HTML, CSS, JavaScript, and WordPress.
- Contributed 8K lines of code to development and refactored SEO, meta, and domain authority, increasing site traffic by 35%.

Operational Supervisor

Six Flags – Kings Dominion

February 2023 – August 2026

- Managed ride operation division teams of up to +70 associates and dozens of large attractions and heavy machinery.
- Received six company-wide awards for reducing coaster operational downtime by 22.8%.

PROJECTS

National Cyber League Tournament Placement

October 2025

- Placed in the top 25% nationally in the NCL Fall 2025 Individual Game Standard Tournament Bracket, solving challenges across 5 domains including web exploitation, cryptography, and forensics categories.
- Analyzed web applications, binaries, logs, and PCAPs using Burp Suite, Ghidra, CyberChef, Wireshark, and Python, applying request manipulation, reverse engineering, regex-based log parsing, and network traffic analysis techniques.
- Achieved a perfect score in Digital Forensics, developing reusable Python parsing workflows and Wireshark methodology.

Vulnerable Active Directory Lab Build

May 2026 – August 2026

- Built a 4-VM Active Directory home-lab in VirtualBox consisting of a Windows Server 2019 Domain Controller, 2 Windows 10 workstations, and a Kali Linux attack host, configuring OUs, user groups, service accounts, and DNS in an isolated network.
- Identified and manually exploited 3 distinct AD attack paths using BloodHound, SharpHound, Impacket, Rubeus, PowerView, and Mimikatz, leveraging Kerberoasting, unconstrained delegation abuse, and GPO privilege escalation.
- Validated 100% of discovered attack paths through both BloodHound graph analysis and live exploitation, then documented remediation guidance and a reproducible build guide that was independently used by 2 classmates to recreate the lab environment.

Mini SIEM + SOC Build

September 2026 – Present

- Engineered an end-to-end Mini SIEM/SOC pipeline in Python that collected and normalized Windows, Linux, and firewall telemetry into a centralized SQLite event store, processing 1,150+ events and 16 simulated attack campaigns.
- Developed 8 independent detection rules mapped to MITRE ATT&CK covering brute-force authentication, suspicious PowerShell, port scanning, impossible logins, privilege escalation, persistence, and administrator account creation.
- Evaluated detection performance against an independent ground-truth dataset, achieving 100% campaign recall, 97% alert precision, and 98% F1.

TECHNICAL SKILLS

Offensive Security: Active Directory, Burp Suite, Metasploit, Nmap, Cobalt Strike, BloodHound, OWASP ZAP

Defensive & SOC: Splunk, CrowdStrike Falcon, Wireshark, Sigma Rules, MITRE ATT&CK, Elastic Stack, Cisco Packet Tracer

Cloud & Infrastructure: AWS IAM, Microsoft Entra ID (Azure AD), Google Cloud, Terraform

Languages: Python, Bash, PowerShell, C++, Rust, JavaScript, SQL, HTML, CSS, PHP, Java, Node.js, Bootstrap, REST API, JSON

ACTIVE CERTIFICATIONS

- NSE 1, NSE 2, NSE 3 (Fortinet Cybersecurity & FortiGate Operator)